

A 130.7-mm² 2-Layer 32-Gb ReRAM Memory Device in 24-nm Technology

Tz-yi Liu, *Member, IEEE*, Tian Hong Yan, *Member, IEEE*, Roy Scheuerlein, Yingchang Chen, Jeffrey KoonYee Lee, Gopinath Balakrishnan, Gordon Yee, Henry Zhang, Alex Yap, Jingwen Ouyang, Takahiko Sasaki, Ali Al-Shamma, *Member, IEEE*, Chinyu Chen, Mayank Gupta, Greg Hilton, Achal Kathuria, Vincent Lai, Masahide Matsumoto, Anurag Nigam, Anil Pai, Jayesh Pakhale, Chang Hua Siau, Xiaoxia Wu, Yibo Yin, Nicolas Nagel, Yoichiro Tanaka, Masaaki Higashitani, Tim Minvielle, Chandu Gorla, Takayuki Tsukamoto, Takeshi Yamaguchi, Mutsumi Okajima, Takayuki Okamura, Satoru Takase, Hirofumi Inoue, and Luca Fasoli, *Senior Member, IEEE*

Abstract—A 32-Gb ReRAM test chip has been developed in a 24-nm process, with a diode as the selection device and metal oxide as the switching element. The memory array is constructed with cross-point architecture to allow multiple memory layers stacked above the supporting circuitry and minimize the circuit area overhead. Die efficiency is further improved by sharing wordlines and bitlines between adjacent blocks. As the number of sense amplifiers under the memory array is limited, a pipelined array control scheme is adopted to compensate the performance impact while utilizing the fast switching time of ReRAM cells. With the chip current consumption being dominated by the array leakage and sensitive to array bias and operating conditions, a charge pump stage control scheme is introduced to dynamically adapt to the operating conditions for optimal power consumption. Smart Read during sensing and leakage current compensation scheme during programming are applied to the large-block architecture and achieve a chip density that is several orders of magnitude higher than prior ReRAM developments.

Index Terms—Charge pump, cross-point, current compliance, leakage current compensation, multiple-layer, nonvolatile memory, ReRAM, sneak path, 3-D architecture.

I. INTRODUCTION

AS NAND Flash memory continues to scale and provides high-density, low-cost solution, Resistive RAM (ReRAM) has been considered as one of the potential technologies for next-generation nonvolatile memory, given its fast access speed, high reliability, and multilevel capability [1]–[3].

Manuscript received April 26, 2013; revised July 12, 2013; accepted August 04, 2013. Date of publication September 20, 2013; date of current version December 20, 2013. This paper was approved by Guest Editor Leland Chang.

T. Liu, T. H. Yan, R. Scheuerlein, Y. Chen, J. K. Lee, G. Balakrishnan, G. Yee, H. Zhang, A. Yap, J. Ouyang, A. Al-Shamma, C. Chen, M. Gupta, G. Hilton, A. Kathuria, V. Lai, A. Nigam, A. Pai, J. Pakhale, C. H. Siau, X. Wu, Y. Yin, M. Higashitani, T. Minvielle, C. Gorla, and L. Fasoli are with SanDisk Corporation, Milpitas, CA 95035-7933 USA (e-mail: Tz-Yi.Liu@sandisk.com).

T. Sasaki and S. Takase are with Toshiba Corporation, Yokohama 247-8585, Japan.

M. Matsumoto is with SanDisk Corporation, Yokohama 247-8585, Japan.

N. Nagel and Y. Tanaka are with SanDisk Corporation, Yokkaichi 512-8550, Japan.

T. Tsukamoto, T. Yamaguchi, M. Okajima, T. Okamura, and H. Inoue are with Toshiba Corporation, Yokkaichi 512-8550, Japan.

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Digital Object Identifier 10.1109/JSSC.2013.2280296

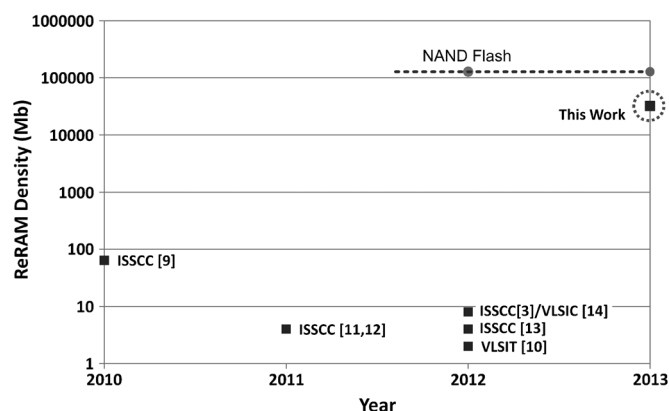


Fig. 1. ReRAM development trend.

These characteristics have been demonstrated or evaluated to be promising for high-density storage application [1], mobile application [4], or as a complementary memory to boost the system performance such as SSD [5], [6] and FPGA [7]. With ReRAM's simple two-terminal cell structure, 3-D multiple-layered architectures have been developed to potentially provide the high-density solution. Various megabit test chips and memory macros have been developed to demonstrate the ReRAM capability and 3-D architectures [3], [8]–[14].

ReRAM array is known to have sneak current during memory operations, and, though large unit array size can provide high die efficiency, the leakage current imposes adverse effect for sensing and writing operations. The severity of the impact depends on the memory cell or selector characteristics, and various circuit or biasing techniques have been proposed to minimize the impact [3], [9], [10]. In this paper, a 32-Gb device is realized by cross-point array with array block size of 16 Mb to accommodate the array control circuit, sense amplifier, page buffer, and regulator driver right underneath the memory array. Several circuit schemes are introduced to overcome the impact of the leakage current and IR drop associated with the large array, as well as to manage the chip current consumption. Fig. 1 shows the recent ReRAM developments, and this work achieves a much higher density than the maximum density of 64 Mb reported previously [3], [9]–[14].

This paper is organized as follows. Section II describes the chip architecture. Pipelined array control scheme is introduced in Section III. The array biasing and read/write circuit are

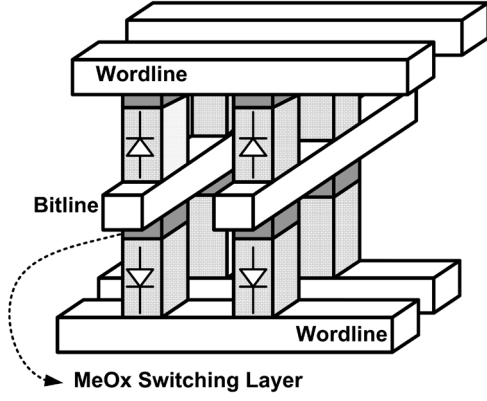


Fig. 2. Memory cell structure.

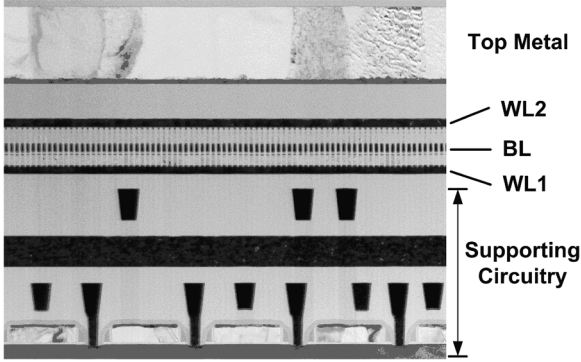


Fig. 3. TEM image of memory array and supporting circuitry.

described in Section IV. Section V describes the charge pump control scheme. The measurement results are presented in Section VI. Finally, the conclusion is given in Section VII.

II. CHIP ARCHITECTURE

A. Array Organization

The memory cell consists of a metal oxide switching element and a diode as the selector. The cell array is formed at the cross points of wordlines and bitlines, as shown in Fig. 2. This chip contains two memory layers with bitlines shared between layers and the available area under the array is fully utilized for the supporting circuitry. Fig. 3 is the TEM image showing the physical construction of the array.

The overall chip organization is described in Fig. 4. The memory core is partitioned into 16 bays, each containing 128 blocks divided into four stripes. The wordline (WL) drivers are placed under the array and the bitline (BL) drivers are located between stripes [15]. The block size is 2 K BLs $\times$ 4 K WLs for each layer, and the adjacent blocks share the same WL and BL drivers to reduce the circuit overhead. Because there is available area under each array block, various circuits can be placed under the array, in addition to the select transistors or final decoder as proposed in previous work [3], [10]. In total, there are 2048 array blocks on the chip, and two different unit block types are distributed across the whole chip to support the memory operations. Fig. 5 shows the chip micrograph and how the two block types are arranged in the chip. Both block types contain column control circuit and regulator driver. In

addition, one of the unit blocks has the sense amplifier, while the other has the array bias control and page register. Each stripe contains one of the two block types.

B. Data Path

Fig. 6 provides the details of the interface between the memory array and the circuit under one bay. The sense amplifier and page register under each bay are shared among the 128 blocks. There are global select buses between each stripe to connect to the corresponding memory blocks. When a pair of memory blocks is selected, the control logic connects the 64 sense amplifiers in a given bay to the global bus GSELB_T[31:0] and GSELB_B[31:0] of the enabled blocks. When another pair of blocks is selected, the connection will be adjusted to connect the sense amplifiers to the new location. GSELB_T[31:0] and GSELB_B[31:0] of the selected blocks are muxed into SELB buses and then connected to the selected BLs through the BL drivers. Within each bay, 64 blocks contain the page register and read/write control logic that communicates to the sense amplifiers in the other 64 blocks. During the operation, page register handles data transfer to and from the periphery, as well as the sense amplifier control.

Because the routing from the sense amplifiers to the selected blocks can go across the bay, the parasitic delay can affect the bandwidth. With limited number of sense amplifiers on the chip (1024 total), address looping is necessary for full-page (2 KB) access. If a page is distributed in one bay, only 64 bits can be accessed simultaneously in each address loop. To fully utilize all the sense amplifiers for maximum performance, each page of data is distributed into two blocks in each bay.

III. PIPELINED ARRAY CONTROL

Beside the limited sense amplifier resources, the IR drop in a large array also puts a constraint on the number of cells that can be written in parallel on the same WL. As a consequence, many write cycles are required for each page operation. Fig. 7(a) describes the array control and data path during write operation. In each write cycle, page register enables the sense amplifiers based on the data at a given address and GSELB bus is charged up by sense amplifiers. The selected BLs are then driven to the target voltage followed by the WL write pulse. Because the GSELB lines span 32 blocks, they are highly capacitive and require some time to charge up and discharge, in addition, the routing distance between the sense amplifier and the selected blocks can be long depending on the location of the sense amplifier and the selected blocks. Considering the fast switching time of the ReRAM cell, the time delay for each address cycle can significantly impact the performance if the data fetching from page register, GSELB charge-up and BL enabling phases are executed in series, as shown in Fig. 7(b).

To address the overhead from the GSELB and BL charge-up and discharge time, a pipelined WL/BL control scheme [16] is shown in Fig. 7(c). Steps (1)–(3) noted in the figure explain the operation. The write pulse is applied by the WL and as the selected cells driven from one side of the block are being written, the data for the next write cycle is read from the page buffer and the sense amplifiers charge up GSELB lines on the other side of the block (2). Once the WL pulse ends, the BLs driven from the other side are charged up quickly for the following write pulse

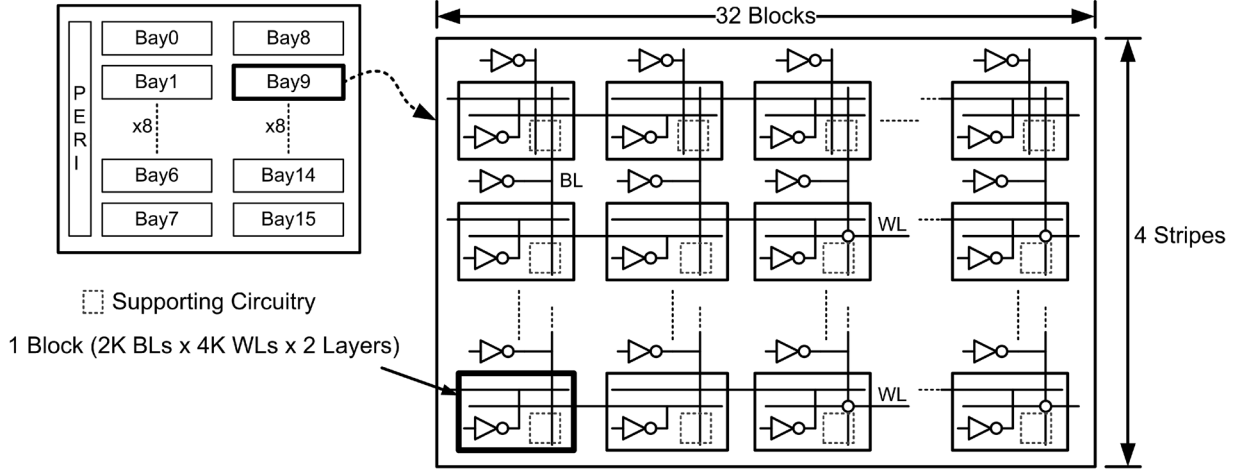


Fig. 4. Array organization.

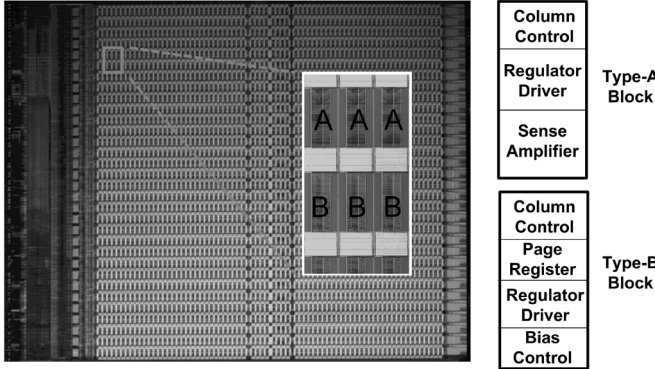


Fig. 5. Chip microphotograph and circuit distribution under array.

(3). The time to fetch the next set of data and charge up the GSELB lines is hidden in this scheme. Compared with the non-pipelined flow, this approach can improve the write throughput by up to 40%.

IV. ARRAY OPERATION

A. Array Biasing in Read Operation

Fig. 8(a) shows the array biasing during the read operation for an $M \times N$ array. The selected BLs and unselected WLs are biased at the read voltage (V_{read}) while the selected WL and unselected BLs are biased at unselected BL bias (V_{UB}). The selected cells (S cells) are biased between V_{read} and V_{UB} voltage level. With this biasing scheme, the half-selected cells along the selected BLs (F cells) and the half-selected cells on the selected WL (H cells) are biased at almost 0 V. The unselected cells (U cells) are reverse biased between V_{read} and V_{UB} , and they conduct little current due to low bias level. Although the H and F cells are targeted to have zero bias, with the parasitic resistance in a large array, the voltage offset from the power supply grid and the circuitry can create nonzero bias across those cells. As shown in Fig. 8(b), the selected BL bias is at target V_{read} level but the unselected WL bias can be at different voltage V_{read}' due to the offset mentioned above. In this case, F cells will be biased at $V_{read} - V_{read}'$ instead of 0 V.

Compared with the selector-less array where self-rectifying cell with high nonlinear factor is needed [17], [18] to avoid half-selected cell leakage limiting the array size and diluting the sensing signal [9], diode selector provides many orders of magnitude on the current ratio between S cells and F/H cells such that even with the bias offset in a large array block, the F/H-cell leakage has little effect on the sensing margin. The large selection ratio also minimizes the impact of data pattern along the accessed BL.

B. Sensing and Smart Read

Since the GSELB bus has large parasitic capacitance, to achieve fast sensing speed, the capacitive load is isolated during sensing. Fig. 9(a) shows the sense amplifier circuit for read operation and the array bias with WLs driven by WL drivers and BLs driven by sense amplifiers and BL drivers. The clamp transistor M1 biases the GSELB bus and the BL at the target read voltage (V_{read}) while isolating the parasitic loading on GSELB from the low-capacitance sense node V_{sense} . The cell current is then conveyed to node V_{sense} and produces sensing signal within $0.9 \mu s$ based on the difference between the reference current I_{ref} and the cell current. The gate voltage (V_{sfg}) of the clamp transistor is common for all the 16 sense amplifiers and generated by a unity-gain buffer with load current of I_{sfg} . The current level of I_{sfg} can be adjusted to favor either reading high-impedance cell or low-impedance cell.

Fig. 9(b) describes the sensing flow. The selected BLs are first pre-charged to V_{read} through switch S1 while the node V_{sense} is biased at V_{sup_sa} by transistor M2. Upon completion of the pre-charge phase, transistor M2 and switch S1 are turned off and I_{ref} is applied to the node V_{sense} . The cell current and I_{ref} are then compared and node V_{sense} swings accordingly. V_{sense} is compared with V_{ref} to generate the output data (sa_rdada). During pre-charge phase, because multiple cells can conduct current, the WL voltage may be elevated due to the WL resistance, which effectively reduces the bias voltage across the cells when the sensing phase starts. During sensing, WL voltage will gradually decrease as the low-impedance cell current is limited by I_{ref} but it may not return to the target level if many conducting cells exist on the WL. The strong-on cells are likely not affected by the elevated WL voltage and can be sensed correctly

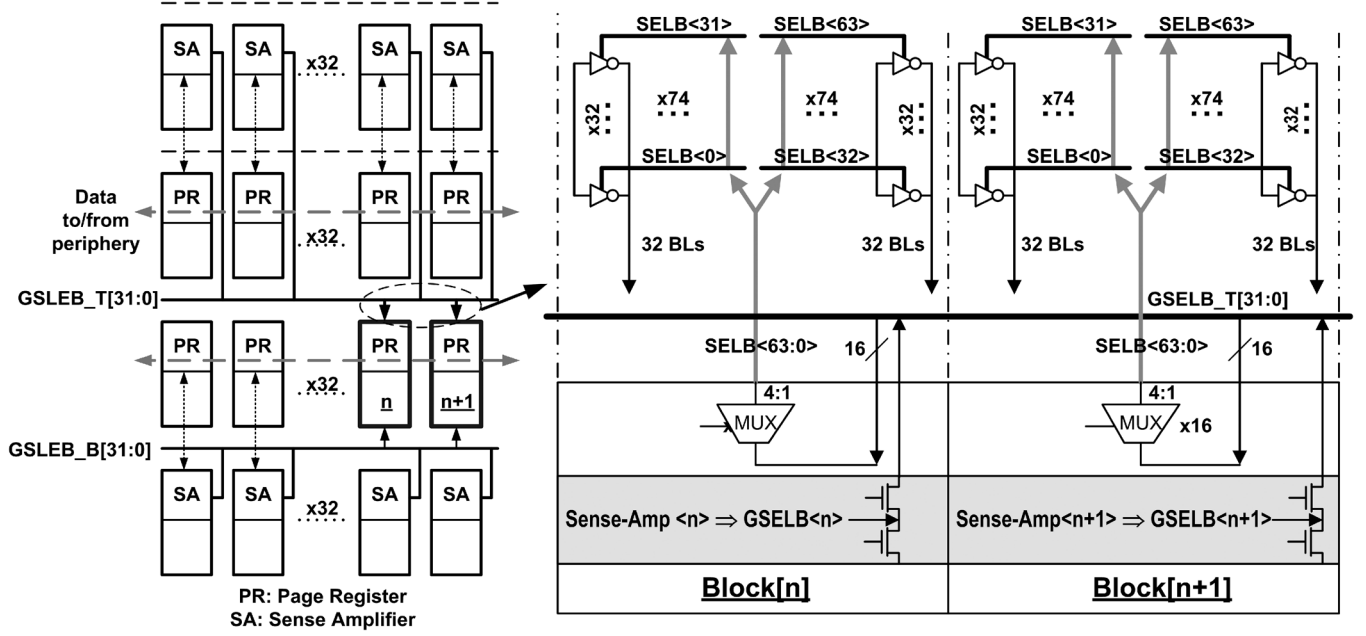


Fig. 6. Shared sense amplifier and page register within a bay.

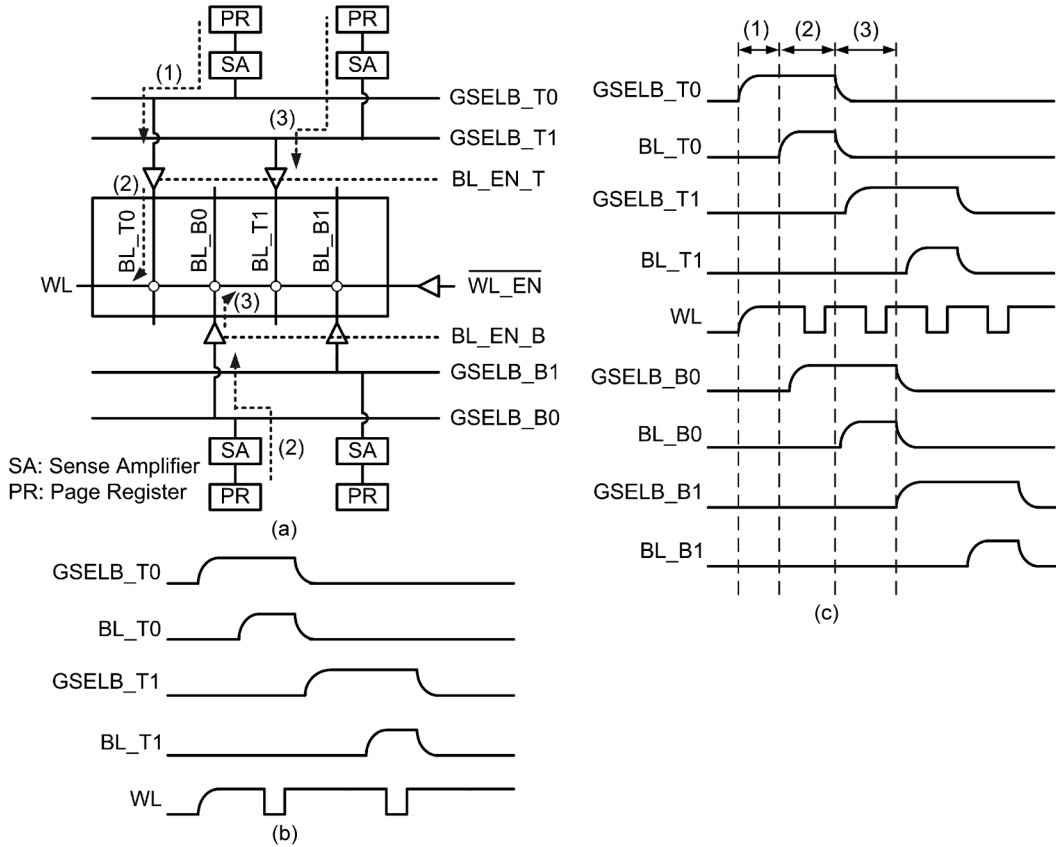


Fig. 7. (a) WL and BL control. (b) Nonpipelined control scheme. (c) Pipelined control scheme.

while the weak cells may fail to trip the sense amplifier if the bias voltage is not recovered enough before the sensing phase ends. Longer sensing time can reduce the WL transient effect but affect the performance. In the program verify scenario, the cells that can conduct current higher than the verify threshold at the target bias may fail in the verify phase and incur unnecessary retry pulses. Additional retry pulses can make the cells more conductive and potentially affect the program capability to switch those cells from the low-resistance state to high-resistance state.

To correctly sense the cells that actually pass the criterion but are read incorrectly due to the WL transient during sensing, necessary retry pulses. Additional retry pulses can make the cells more conductive and potentially affect the program capability to switch those cells from the low-resistance state to high-resistance state.

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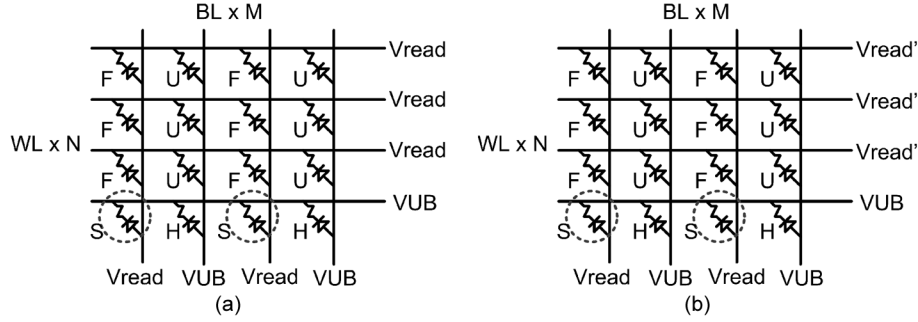


Fig. 8. (a) Array biasing during read operation. (b) Bias offset during read operation.

a Smart Read approach is proposed where the sense amplifier discharges the BL when the corresponding cell conducts current higher than I_{ref} [19]. As the stronger on cells are sensed and turned off actively by the sense amplifiers, the WL potential can recover faster and be closer to the target level for the weak cells conducting more than I_{ref} threshold to be sensed correctly. Fig. 9(c) shows the transient characteristics of various signals with this scheme, where *sa_rdata* state is used to enable the BL discharge signal (*dischg*) at transistor M3 shown in Fig. 9(a)

The proper I_{ref} level is determined by memory cell characterization such that enough on/off window and the lowest failure rate can be obtained. V_{ref} level affects the sensing time as V_{sense} node needs to swing below V_{ref} for an ON cell to be read correctly. Although higher V_{ref} can shorten the sensing time, the noise implication from V_{sup_sa} should be considered.

C. Array Biasing in Program Operation

Fig. 10(a) describes the array biasing during program operation. The selected BLs are driven at the target voltage VWR by sense amplifiers and the selected WL is biased at 0 V. The unselected U cells are reverse biased between VUX and VUB. The H and F cells are forward-biased below the disturb level and can conduct current along the selected BL and WL, which causes IR drop and reduces the voltage across the S cell. The equivalent circuit of the array is shown in Fig. 10(b). VUX and VUB are generated by internal voltage regulators. The U-cell bias level determines the array leakage and contributes to the majority of the chip current consumption due to large number of the unselected cells. Higher VUX and/or lower VUB level can reduce H and F cell current (I_H and I_F) to provide higher voltage and current to the S cell. On the other hand, this increases the U cell bias and results in higher chip I_{cc} . Array bias trade-off needs to be made between the reverse leakage current from U cells and the leakage level of F and H cells to deliver the required voltage and current to the S cells while maintaining reasonable I_{cc} .

With the independent power supplies VUX and VUB controlling the unselected bias, they can be adjusted to modulate the I_F and I_H current to maximize the current delivered to the selected cells. Shown in Fig. 11(a) is the chart of the maximum voltage that can be reached at the selected cell (V_s) in a block versus F-cell bias where the VUB level is kept at a fixed level. Higher V_s can be achieved when F-cell is biased at lower voltage (higher VUX) due to the decreased IR drop on the selected BL, at the cost of U-cell leakage increase. To provide higher V_s with higher VWR, VUX level needs to track VWR to avoid

increasing F-cell leakage and IR drop on the BL. In Fig. 11(b), similar trend is shown for the current that can be delivered to the selected cell (I_s). Also, with a given VWR voltage, there is an optimal bias range where I_s can be maximized, as shown in Fig. 11(c). Each bias condition (A, B, C, D) marked in the chart represents a U-cell bias level with condition D being the highest and condition A being the lowest among the four cases. For a given U-cell bias level, I_s can be maximized by adjusting the F-cell and H-cell bias to optimize the IR drop in the WL and BL paths.

D. Leakage Current Compensation

As previously mentioned, the F-cell leakage can offset the current delivered to the selected cell and affect the programming yield. As more cells are programmed into low-resistance state along the same BL, the leakage increases and is data pattern dependent. To minimize the impact from the leakage current along the selected BL, the sense amplifier samples the leakage current before the write pulse is asserted. As shown in Fig. 12(a), all of the WLs are initially biased at VUX in the sampling phase when sense amplifier samples the leakage level and produce the compensation current (I_{comp}) to be sourced into the BL. WL pulse is then enabled to program the selected cells. BL voltage is regulated based on the target voltage VWR and BL current is mirrored and compared with the current reference (I_{ref}) to monitor the cell state. Without the leakage compensation, when large number of cells along the same BL is programmed, F-cell current can be higher than I_{ref} and cause false detection of the cell state transition when the selected cell is not actually programmed. In addition, to provide current compliance to the selected cell during program operation, the leakage compensation is necessary to minimize the impact of the stray current. Fig. 12(b) shows the relative timing among the selected WL, selected BL, and sense amplifier output current (I_{SA}). At the end of the each BL pre-charge period when all WLs are at VUX, I_{SA} is settled to the F-cell leakage level and the leakage current is stored in a sampling circuit to generate I_{comp} . Initial I_{SA} transient depends on the charge-up scheme. When the program pulse is applied, I_{comp} is sourced into the selected BL while the current from the selected cell (I_{cell}) is monitored. Since the leakage current sampling is performed during the BL pre-charge phase, it does not incur additional overhead.

Table I summarizes the bias conditions for read and write operations.

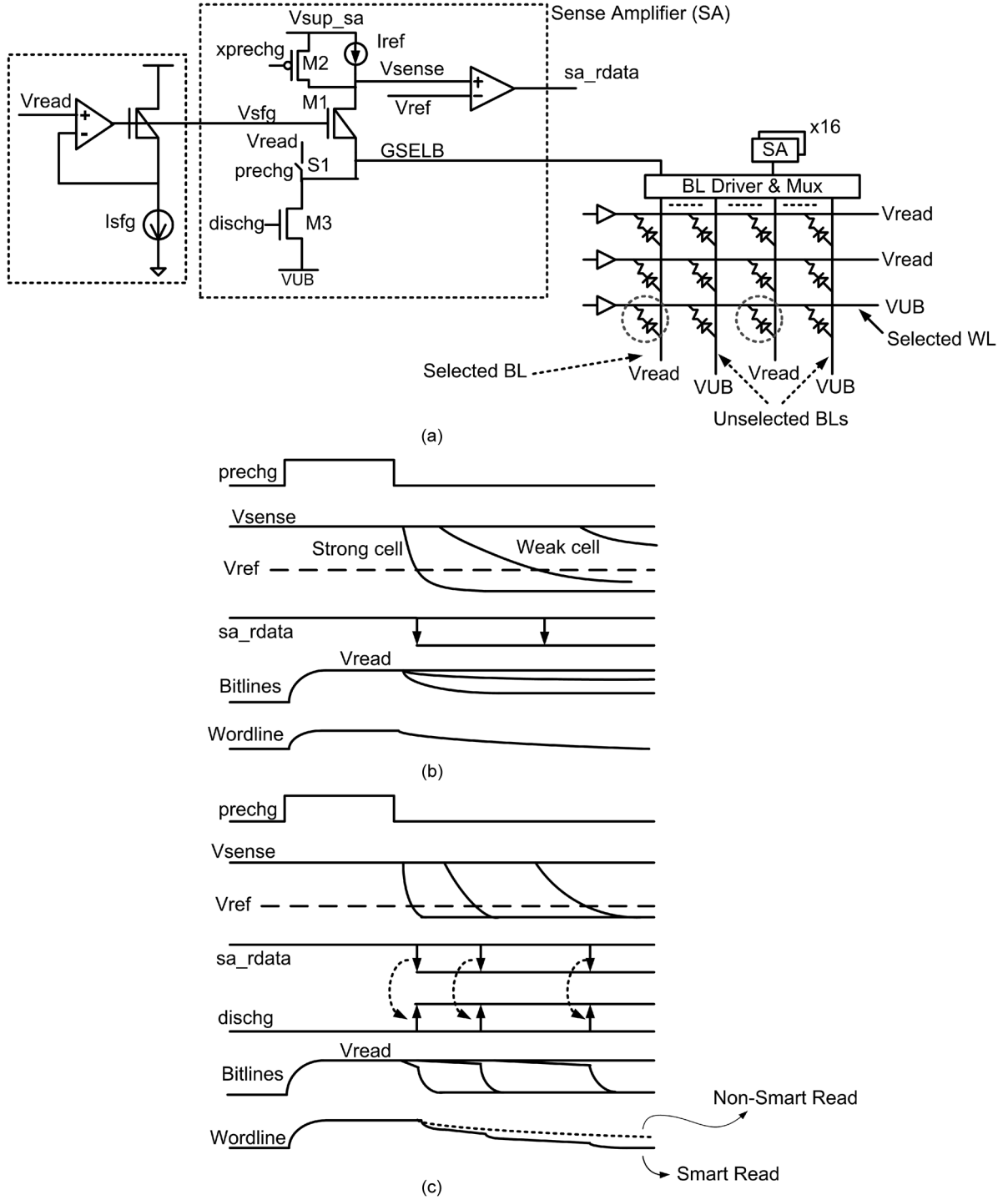


Fig. 9. (a) Sense amplifier circuit. (b) Sensing without detection. (c) Sensing with detection (Smart Read).

V. CHARGE PUMP CONTROL

With large block size, the array leakage contributes a significant portion of the overall chip I_{cc} . As the charge pump is the main voltage source to supply the array bias and the leakage current loading is sensitive to the operating conditions, the charge pump efficiency is important to support wide range of array bias while maintaining reasonable I_{cc} level. For a given output voltage, there is an optimal number of charge pump stages to achieve maximum power efficiency [20], [21]. Fig. 13(a) is the

simulation result of power efficiency vs. output voltage and number of voltage doubler stages. Fig. 13(b) shows the simulation result of the pump clock frequency which is a function of output voltage and the number of stages enabled. When the output voltage target level exceeds the optimal output level of the pump at a given number of stages, the efficiency begins to decrease and the clock frequency increases dramatically, which can be utilized as a criterion to determine the optimal number of stages and limit the efficiency drop. For example, for a three-stage pump to deliver an output voltage beyond the

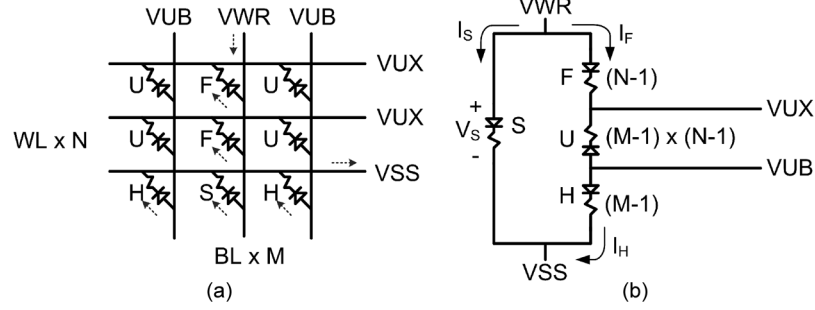


Fig. 10. (a) Array biasing during program operation. (b) Equivalent circuit.

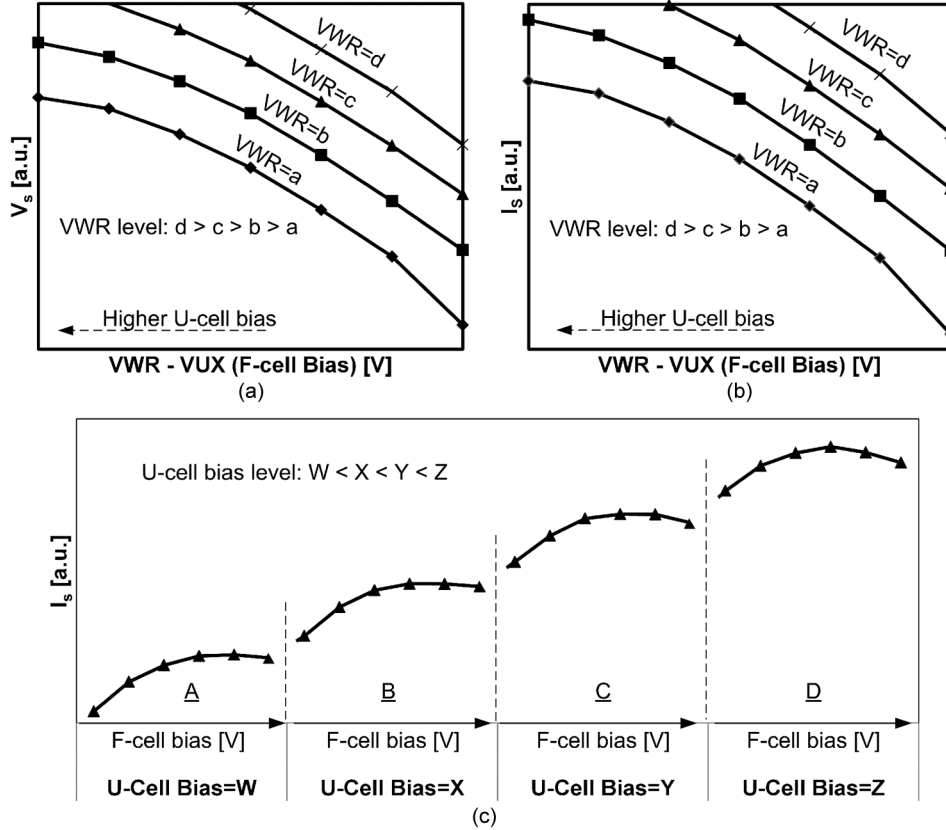
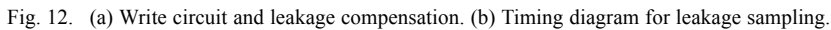


Fig. 11. (a) Maximum selected cell voltage (V_s) with fixed VUB level. (b) Maximum selected cell current (I_s) with fixed VUB level. (c) Maximum selected cell current (I_s) with fixed VWR level. All charts are drawn in linear scale.

optimal point, the clock frequency will be around or more than $2\times$ of the clock frequency of the four-stage configuration. If the pump control can detect the large frequency change and adjust the configuration to four stages, the efficiency drop can be limited. With this self-adjusting approach, the power efficiency can be maintained at a relatively consistent level from one operating condition to another, or provide a baseline configuration when the program operation starts.

Fig. 14 describes the details of the charge pump stage control scheme where a small monitor pump is introduced. At the beginning, both main pump and monitor pump are started with the same number of stages. The test current I_{test} is then changed such that the clock frequency of the monitor pump is locked to the main pump clock frequency. This is the lock-in phase, as shown in Fig. 14(a). The I_{test} level and the size of the monitor pump are chosen such that the added active current is insignificant compared with the total chip current consumption.

The next step is illustrated in Fig. 14(b). Once the monitor pump clock locks in the main pump clock, the control circuit will increase the number of stages for monitor pump by one. As a consequence, the clock frequency of the monitor pump will decrease. The clocks of the two pumps are then compared, and, if the monitor pump clock frequency is less than half of the main pump clock frequency, it means adding one stage is more efficient and the control circuit will enable one more stage for the main pump. Similarly, the monitor pump configuration will also be reduced by 1 stage, which will result in faster clock. If the monitor pump clock frequency is less than twice of the main pump clock frequency, it indicates fewer stages is more efficient and the number of stages for main pump is reduced by 1, as shown in Fig. 14(c). The lock-in and frequency comparison process with different monitor pump configurations continues throughout the operation and the main pump configuration is changed only when the frequency comparison meets the criteria.



	Read	Write
Selected WL	VUB	0V
Unselected WL	Vread	VUX
Selected BL	Vread	VWR
Unselected BL	VUB	VUB

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VI. MEASUREMENT RESULTS

With the leakage current compensation, programming large number of cells along the BL is feasible and can achieve good yield. The passing bit count per page in page-by-page programming is shown in Fig. 17. Without the automatic leakage compensation, the yield degrades quickly as more bits

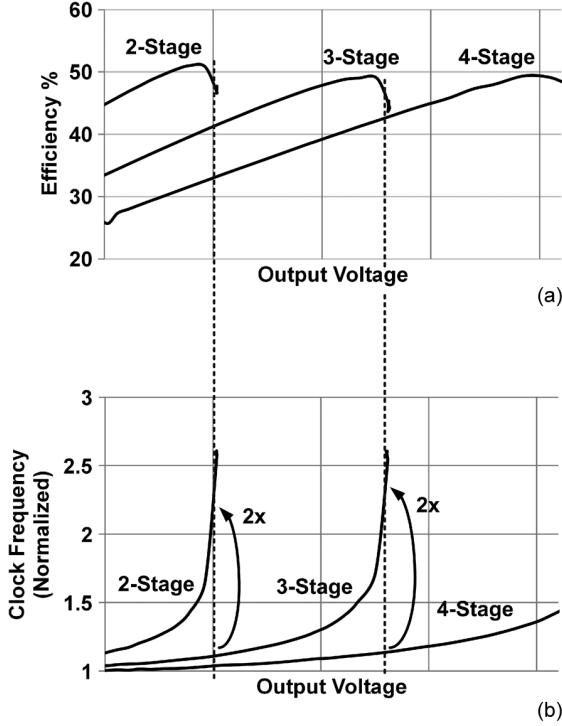


Fig. 13. (a) Charge pump power efficiency versus output voltage and number of stages. (b) Charge pump clock frequency versus output voltage and number of stages.

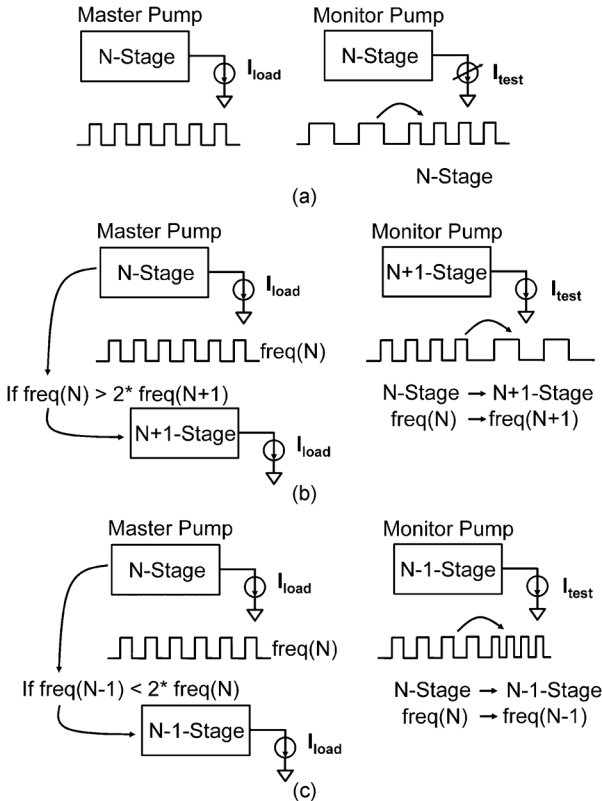


Fig. 14. Dynamic charge pump stage control. (a) Lock-in phase. (b) Clock frequency comparison with added stage in monitor pump. (c) Clock frequency comparison with one stage removed from monitor pump.

are programmed along the BL. With the proposed compensation scheme, consistent yield can be achieved throughout the operations.

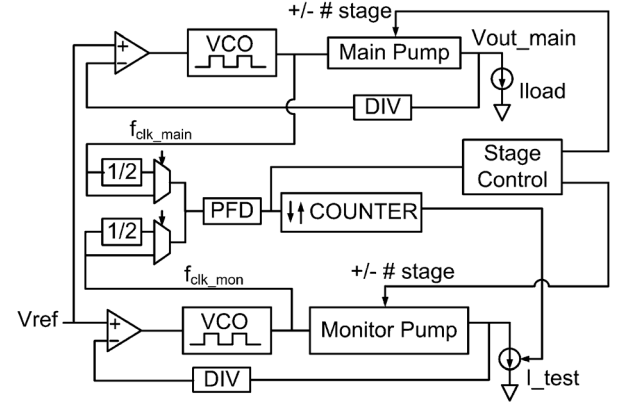


Fig. 15. Charge pump control system.

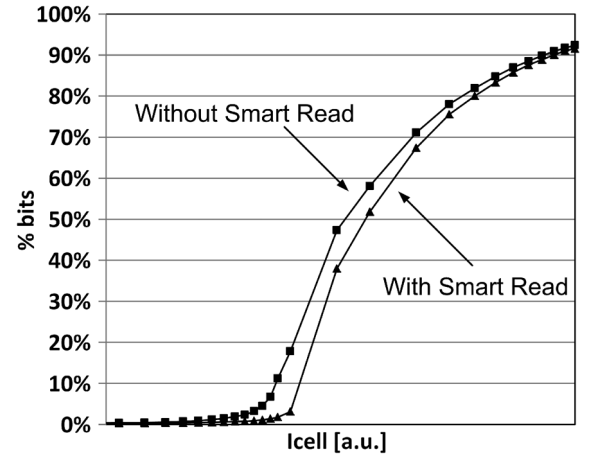


Fig. 16. Measured cell current distribution showing Smart Read improves read current level during sensing.

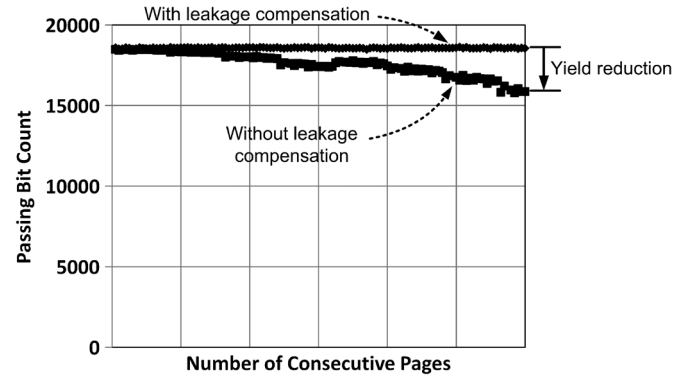


Fig. 17. Measured programming yields improvement with leakage current sampling.

The operation of the proposed charge pump control is shown in Fig. 18. The stage configuration of the monitor pump is constantly changed to evaluate the clock frequency. The main pump stage configuration normally stays constant. Fig. 18(a) shows the number of the stage for the main pump is increased when the load current increases. Fig. 18(b) shows that as the V_{cc} level is increased, the control logic reduces one stage for the main pump. The main pump configuration is adjusted only when the

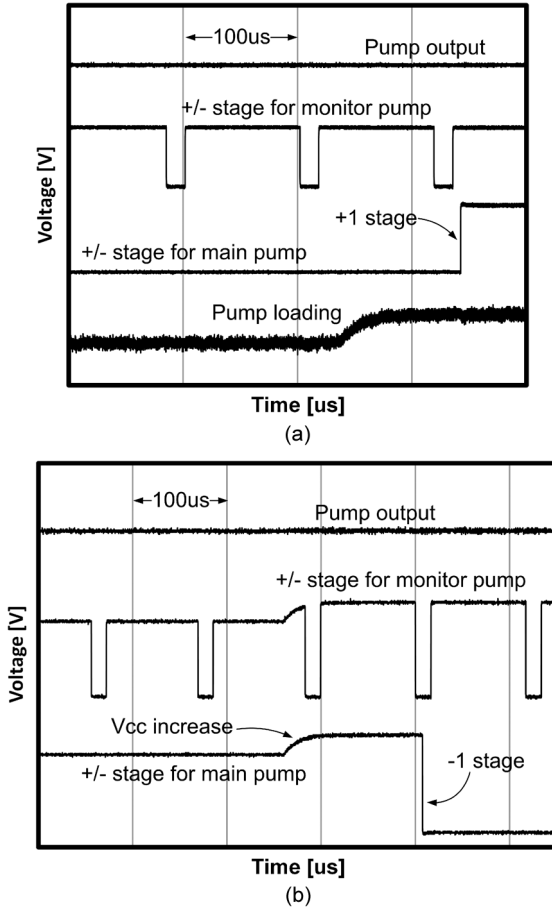


Fig. 18. Measured waveforms with dynamic charge pump stage control. (a) The main pump configuration is increased by one stage after the load current is increased. (b) The main pump configuration is decreased by one stage after Vcc level is increased.

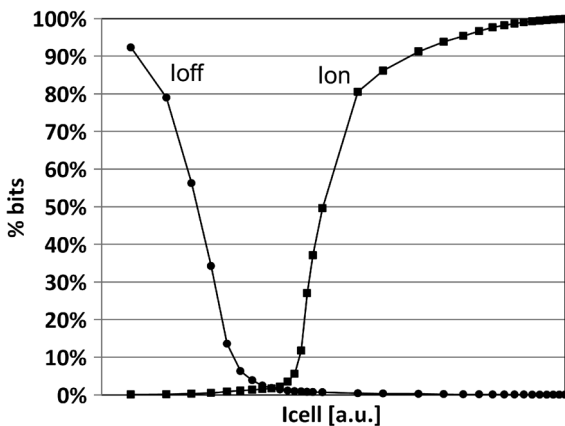


Fig. 19. Measured cell current distribution from one 2-KB page.

frequency comparison result meet the decision criteria. The experimental result shows that up to 25% of I_{cc} saving from the pump can be achieved with this control scheme.

Lastly, Fig. 19 shows a sample cell current distribution from a full 2-KB page.

TABLE II
DEVICE FEATURES

Density	32Gb
Die Size	130.7mm ²
Interface	NAND-Compatible
Page Size	2KB
Read Latency	40us
Write Latency	230us

VII. CONCLUSION

A 32-Gb ReRAM test chip is developed in 24-nm technology with cross-point architecture, representing a major increase in density compared with prior ReRAM developments. Large array block size is adopted in this chip to allow most of the array supporting circuitry to be placed under the array to minimize the die area overhead. The performance impact due to shared sense amplifier and array control circuit is reduced by up to 40% with the pipelined array control scheme. Smart Read approach allows the memory cells to be read at a more accurate bias level and improves the sensing capability for the weak cells. The effect of leakage current caused by the sneak path in a large array is mitigated by applying sampled leakage current to the BL during programming operation and consistent yield is achieved. A dynamic charge pump control scheme is introduced to optimize the charge pump configuration for power consumption based on the operating conditions. Table II. summarizes the device features. With the potential expandability of cross-point architecture, this chip provides a test vehicle to evaluate the memory cells, array architecture, and circuit solutions for high-density ReRAM device.

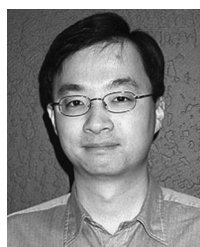
ACKNOWLEDGMENT

The authors would like to thank S. Mehrotra, K. Quader, M. Mofidi, R. Shrivastava, K. Kobayashi, Y. Naruke, S. Mori, N. Hayasaka, T. Hara, and the Process, Integration, Device, Evaluation, Test, Layout and CAD teams from SanDisk and Toshiba for supporting the ReRAM development.

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Tz-yi Liu (M'92) received the B.S. degree from National Taiwan University, Taipei, Taiwan, in 1994, and the M.S. degree from Stanford University, Stanford, CA, USA, in 1998, both in electrical engineering.

From 1998 to 2002, he was with Micron Technology, Inc., where he worked on memory core, data path, analog circuit, and DC–DC converter for system power reduction. In 2002, he joined Matrix Semiconductor (acquired by SanDisk in 2006), where he was engaged in the design of 3-D nonvolatile memory devices, including architecture development, circuit implementation, place and route of the on-chip state machine, as well as managing the physical design integration. Since 2008, he has been leading the ReRAM design effort. He holds one Taiwan patent and more than ten U.S. patents issued or pending and has authored or coauthored five technical publications.



Tian Hong Yan (M'95) received the B.S. degree in electromechanical engineering from Shanghai University of Technology, Shanghai, China, in 1990, and the M.S. degree in electrical engineering from San Jose State University, San Jose, CA, USA, in 1993.

From 1993 to 2000, he was with Mosel/Vitelic Corporation, working on various projects from analog-storage with E2PROM to Split-gate NOR-type Flash-memory products. From 2000 to 2006, he was with Integrated Memory Technology Corporation, where he was the leading designer on array core, the data-path, and embedded controller for multiple NAND-interfaced NOR-type Flash-memory test-chip and products. In early 2006, he joined SanDisk, and worked on the design of 3-D OTP products. He is now a Senior Manager focusing on array core design for 3-D ReRAM products.



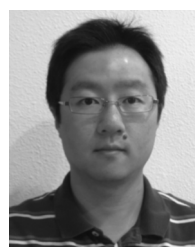
Roy Scheuerlein received the B.S. degree in physics and Ph.D. degree in electrical engineering from Stanford University, Stanford, CA, USA, focusing on semiconductor memory.

He held a variety of engineering positions with IBM, where he focused on leading-edge memory technologies moving them from the research stage to product introduction including trench DRAM memories; high speed DRAMS; and embedded Flash memory. He developed the array architecture for high density and high performance (3 ns) MRAM memory and designed and presented the first MRAM memory development vehicle at the 2000 ISSCC. At the startup company, Matrix Semiconductor he developed the architecture for the first commercially successful 3-D OTP memory product. Since the acquisition of Matrix Semiconductor Company by SanDisk, he has continued to develop 3-D memory. His current area of interest is 3-D ReRAM memory design and technology for the post NAND era.



Yingchang Chen received the B.S. degree in control engineering from National Chiao Tung University, Hsinchu, Taiwan, in 1992, and the M.S. degree in electrical engineering from National Taiwan University, Taipei, Taiwan, in 1994.

From 1997 to 2001, he was with Integrated Telecom Express Inc., Santa Clara, CA, USA, as an Analog Circuit Designer, working on the analog-front-end for ADSL, including ADC, DAC, amplifiers, and filters. In 2001, he joined Redswitch Inc., later acquired by Agilent Technology, Santa Clara, as a Senior Circuit Design Engineer, to work on the high-speed I/O circuits, including PLL and SERDES. From 2006 to 2007, he was with Mober Semiconductor, Fremont, CA, USA, as a Design Manager to design and lead the projects of GSM transceiver IC. In late 2007, he joined SanDisk Corporation, Milpitas, CA, USA, where he has been engaged in the 3-D memory development.



Jeffrey KoonYee Lee received the B.S.E.E. and M.S.E.E. degrees from Hong Kong University of Science and Technology, Hong Kong, in 2003 and 2005, respectively, and the M.S.E.E. degree from the University of Southern California, Los Angeles, CA, USA, in 2007.

He is currently a Member of the circuit design team at SanDisk, Milpitas, CA, USA, working on three-dimensional memory design.



Gopinath Balakrishnan received the B.Eng. degree in electronics and communication engineering from University of Madras, Chennai, India, in 2002, and the M.S. degree in electrical engineering from University of Nevada, Las Vegas, NV, USA, in 2005.

From 2006 to 2007, he was a Member of the Product Engineering Group, Texas Instruments Inc., Dallas, TX, USA, where he was primarily involved with debugging the new Silicon. In 2007, he joined SanDisk Corporation, Milpitas, CA, USA, as a Circuit Design Engineer, where he is engaged in architecture definition, data-path design, and analog circuit design including temperature sensor, voltage regulator and reference generators, and debugging new silicon for 3-D memory. He has seven U.S. patents issued or pending.



Gordon Yee received the B.S. degree in electrical engineering and computer sciences from the University of California, Berkeley, CA, USA, in 1991, and the M.S. degree in electrical engineering from the University of California, Los Angeles, CA, USA, in 1993.

In 1993, he joined Cirrus Logic, Fremont, CA, USA, where he was involved with the design of DACs and SRAMs for graphics chips. In 1996, he joined Silicon Graphics, Mountain View, CA, USA, where he designed floating-point multipliers and adders for MIPS microprocessors and a video input engine for use in a visual workstation. In 2001, he joined Broadcom, San Jose, CA, USA, working on the design and integration of DDR memory controllers and ADSL routers. In 2003, he joined Matrix Semiconductor, Santa Clara, CA, USA, where he was responsible for the verification of several one-time-programmable three-dimensional memories. In 2006, Matrix Semiconductor was acquired by SanDisk, Milpitas, CA, USA, where he is currently a Senior Manager involved with logic design and verification as well as product development for the memory design group.

Henry Zhang, photograph and biography not available at the time of publication.



Alex Yap received the B.S.E.E (with Highest Distinction) and M.S.E.E. degrees from Purdue University, West Lafayette, IN, USA, in 1988 and 1989, respectively.

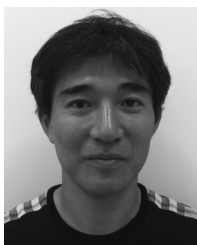
Upon graduation, he spent the next 24 years in Silicon Valley as a Logic Architect and ASIC Designer in the fields of printer controller, CPU, 3-D graphics, video encoder/decoder, memory design, and power management IC. He has worked for startup companies including SMOS systems Inc., Oak Technology Inc., Transmeta Corporation, T-RAM Inc., Richcore

Inc., and Fystorm Inc. He joined SanDisk Corporation, Milpitas, CA, USA, back in 2006 as a Member of the 3D Logic and Verification team developing 3-D memory controller.



Jingwen Ouyang received the B.S. degree and M.Eng. degree in electrical engineering from the Massachusetts Institute of Technology, Cambridge, MA, USA, in 2008 and 2009, respectively.

In 2010, she joined SanDisk Corporation, Milpitas, CA, USA, as a Circuit Design Engineer working on 3-D memory design.



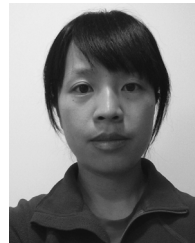
Takahiko Sasaki received the B.S. and M.S. degrees and Ph.D. degree in physics from the University of Tokyo, Tokyo, Japan, in 1998, 2000, and 2004, respectively.

He joined Toshiba Corporation in 2003, where he was engaged in SRAM design development. He is currently engaged in flash memory design development at the Center for Semiconductor Research and Development, Toshiba Corporation, Kawasaki, Japan.



Ali Al-Shamma (M'96) received the B.S. and M.S. degrees in electrical engineering from Santa Clara University, Santa Clara, CA, USA, in 1993 and 1996, respectively.

He was with AMD's Nonvolatile Memory Division from 1997 to 2000, where he worked on NOR ultralow-voltage flash memory design. In 2001, he joined Matrix Semiconductor where he worked on developing 3-D memory products. In 2006, he joined SanDisk Corporation, Milpitas, CA, USA, through its acquisition of Matrix Semiconductor. He has been involved in 3-D memory sub-block design, architectural definition and most recently as a Project Manager for a next generation 3-D memory test chip. He holds more than 20 patents.



Chinyu Chen received the B.S.E.E. degree from Tsinghua University, Beijing, China, in 2004, and the M.S.C.E. degree from Syracuse University, Syracuse, NY, USA, in 2007.

She is currently a Member of the Logic Design Team of SanDisk Corporation, Milpitas, CA, USA, verifying 3-D memory designs.



Mayank Gupta received the B.S. degree from the University of Madhya Pradesh, India, in 2005, and the M.S. degree from the University of Southern California, Los Angeles, CA, USA, in 2006, both in electrical engineering.

In 2007, he joined Marvell Semiconductors, Santa Clara, CA, USA, as an ASIC Design Engineer, where he primarily worked on the PHY design for wireless SOCs. In 2011, he joined SanDisk Corporation, Milpitas, CA, USA, as a Senior Design Engineer. He is currently working on ASIC Design for 3-D memory controllers.



Greg Hilton received double-major B.S. degrees in electrical engineering and software engineering from Cogswell Polytechnical College, Sunnyvale, CA, USA, in 2006, and the M.S. degree in software engineering from Carnegie Mellon University, Pittsburgh, PA, USA, in 2010.

He joined SanDisk Corporation, Milpitas, CA, USA, in 2005, as an Intern for the ASIC team working on power on reset testing, IO characterization, and mixed-mode verification. In 2007, he joined the Memory Design team working on 3-D memory projects. He currently a Senior Verification Engineer working on all aspects of verification: improving coverage, simulation performance, and methodology.

Achal Kathuria, photograph and biography not available at the time of publication.

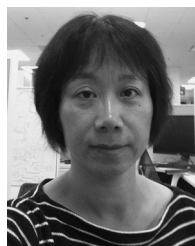
Vincent Lai, photograph and biography not available at the time of publication.



Masahide Matsumoto received the B.S. degree in electrical engineering from Chuo University, Tokyo, Japan, in 1984.

He worked on DRAM product design with Texas Instruments Japan from 1984 to 1998. From 1998 to 2002, he continued working on mixed-signal product design and led I/O design team for ASIC product. From 2002 to 2007, he was a Circuit Design Engineer with UMC Japan where he worked on mixed-signal IP design such as high-speed I/O and data converters. In 2007, he joined SanDisk Japan, where he has been

working on analog circuit design of NAND flash memory and 3-D memory. He holds one U.S. patent.



Xiaoxia Wu received the B.S. degree from the University of South China Technology, China, and the M.S. degree from San Jose State University, San Jose, CA, USA.

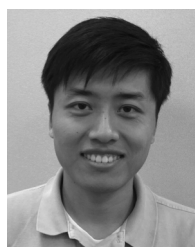
Since 1997, she has worked on NOR, NAND Flash memory products with Hynix Semiconductor and Micron Technology. She is currently working on 3D ReRAM memory design with SanDisk Corporation.



Anurag Nigam received the B.S. degree from Indian Institute of Technology, Kanpur, India, in 2006, and M.S. degree from the University of Virginia, Charlottesville, VA, USA, in 2010, both in electrical engineering. His M.S. research was focused on bit-cell modeling and design for Spin-Transfer Torque (STT-RAM) memory.

From 2006 to 2008, he was a Software Engineer with United Online and Infinera. In 2011, he joined SanDisk Corporation, Milpitas, CA, USA, as a Circuit Design Engineer. He is currently working on

sense amplifier circuits for 3-D memory. He is an author or coauthor of five technical publications.



Yibo Yin received the B.S.E.E. degree from Beihang University, Beijing, China, in 2008, and the M.S.E.E. degree from the University of Southern California, Los Angeles, CA, USA, in 2011.

He is currently a member of the Logic Design Team with Sandisk Corporation, Milpitas, CA, USA, designing and verifying 3-D memory devices.



Anil Pai received the B.Tech. degree in electronics and communication engineering from the National Institute of Technology, Trichy, India, in 2008, and the M.S. degree in electrical engineering from the University of Southern California, Los Angeles, CA, USA, in 2010.

After his graduation, he joined SanDisk Corporation, Milpitas, CA, USA, where as a Circuit Design Engineer, he was involved with the design and development of NAND flash memory and 3-D memory.



Nicolas Nagel received the Ph.D. degree in physics from the Max-Planck-Institute for Solid States, Stuttgart, Germany, in 1993.

He joined Sony, Atsugi, Japan, developing FeRAM as a Process Engineer. Then, he joined Infineon, developing DRAM with high- k dielectrics first in Munich, Germany, and then with the DRAM Alliance IBM—Toshiba—Infineon in East Fishkill, NY, USA. In 2001, he joined the FeRAM Development Alliance Toshiba—Infineon in Yokohama, Japan, as an Integration Manager. Back in Dresden,

Germany, he worked on Flash development, leading a Twin-Flash (NR0M) and NAND—Flash technology project. From 2008 he is working for Sandisk in the Toshiba—Sandisk Alliance in Yokkaichi, Japan, leading the 3-D ReRAM development technology project.



Jayesh Pakhale received the B.S. degree in electronics and communication engineering from Nirma Institute of Technology, India, in 2005, and the M.S. degree in electrical engineering from San Diego State University, San Diego, CA, USA, in 2009.

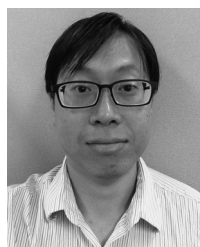
He started his career as a Hardware Design Engineer with Entity Solution Ltd, India, in 2005. In 2011, he joined SanDisk Corporation, Milpitas, CA, USA, as a Verification Engineer. He is currently a Member of the Logic Design and Verification Team, developing and verifying 3-D memory designs.



Yoichiro Tanaka received the B.S. degree from the University of California, Berkeley, CA, USA, in 1989, and the Ph.D. degree from Nagoya University, Nagoya, Japan, in 2004.

He started his career at Applied Materials Japan, Narita, Japan, and later joined Applied Materials, Santa Clara, CA, USA, where he led multiple projects in semiconductor metallization technology up to 2006. He played a key role in the development and introduction of industry's first ionized physical vapor deposition technology to the quarter micron

device generation. Also during this period, he has taken numerous overseas assignments in the development of BEOL thin films and integration technology with various fabrications around the world. He joined SanDisk Corporation, Milpitas, CA, USA, working on 3-D memory division in 2006, after the acquisition of Matrix Semiconductor. He joined the 3D Process Technology Team, where he contributed to the various thin films development for OTP device at 45-nm generation. Since 2008, he has been taking long term assignment in Toshiba Yokkaichi Fab, Mie, Japan, where SanDisk/Toshiba conducts joint development of 3-D ReRAM device. He leads a team of material and process integration engineers in the development of 2×nm generation ReRAM device.



Chang Hua Siau received the B.S. degree in electrical engineering from Iowa State University, Ames IA, in 1998. He was with the Nonvolatile Memory Division of Micron Technology Inc., Santa Clara, CA from 1998 to 2005, involved in the design of NOR Flash memory. In 2005, he joined Unity Semiconductor Corp., Sunnyvale CA where he was engaged in the definition and implementation of cross point non-volatile memory array architecture, sensing scheme, and supporting circuits. In 2011, he joined SanDisk Corp., Milpitas CA, working in 3D

semiconductor memory design team. He holds over 27 U.S. patents issued or pending and has published 3 technical papers.



Masaaki Higashitani was born in Hyogo, Japan, in 1965. He received the B.S. degree from Yamagata University, Yamagata, Japan, in 1987, and the M.S. degree from Kanazawa University, Kanazawa, Japan in 1989, both in physics.

In 1989, he joined Fujitsu, Kawasaki, Japan, where he worked on device/process integration for DRAM and nonvolatile memories. In 2001, he joined SanDisk Corporation, Milpitas, CA, USA, where he is engaged in the development of device/process integration for high-density NAND

Flash memory and ReRAM.

Tim Minvielle received the B.S. degree in physics and Ph.D. degree in electrical engineering from Stanford University, Stanford, CA, USA.

He is an Engineering Director with SanDisk Corporation, Milpitas, CA, USA. He is working on next-generation NAND replacement technologies. Prior to this, he worked at Intel on phase-change memory development and IBM in the hard-drive division.



Chandu Gorla received the B.S. degree in metallurgical engineering from the Indian Institute of Technology, Madras, India, in 1992, and the Ph.D. degree in materials science from Rutgers University, Piscataway, NJ, USA, in 1999.

From 2000 to 2007, he was with Cypress Semiconductor, San Jose, CA, USA, working on developing BiCMOS and RF-CMOS technologies for wireless applications. In 2007, he joined SanDisk Corporation, Milpitas, CA, USA, where he has been involved with FG NAND and 3-D ReRAM device engineering.

Takayuki Tsukamoto received the B.Sc. and D.Sc degrees from the University of Tokyo, Tokyo, Japan, in 1992 and 1997, respectively.

He joined the Research and Development Center, Toshiba Corporation, in 1997. He was a Visiting Scholar with Stanford University, Stanford, CA, in 2005. Since 2007, he has been with Semiconductor Company, Toshiba Corporation, Yokkaichi, Japan.



Takeshi Yamaguchi received the B.S., M.S., and Ph.D. degrees from the University of Tsukuba, Tsukuba, Japan, in 1992, 1994, and 1997, respectively, all in materials science.

He joined the Research and Development Center, Toshiba Corporation, in 1997, where he was engaged in the research on the physics and technology of ferroelectric devices. From 1999 to 2006, he was engaged in the research on the reliability physics and properties of MISFET with high-k gate dielectrics. Since 2007, he has been with the Memory Division,

Toshiba Corporation, Yokkaichi, Japan, where he has been leading the ReRAM memory cell development.



Mutsumi Okajima was born in Aichi, Japan, in 1967. He received the B.E. and M.E. degrees in electronic engineering from Waseda University, Tokyo, Japan, in 1990 and 1992, respectively.

In 1992, he joined Toshiba Corporation, Kawasaki, Japan, where he was engaged in the process integration for DRAM. Since 2002, he has been working on the process integration for NAND flash memories and other nonvolatile memories.



Takayuki Okamura received the B.E. and M.E. degrees in material engineering from Waseda University, Tokyo, Japan, in 1987 and 1989, respectively.

He joined Toshiba Corporation, Kawasaki, Japan, in 1989, where he was engaged in the development of dynamic memory products, including their device and process technologies. From 2006 to 2009, he worked on the development of NAND flash memory and ReRAM devices at Center for Semiconductor Research and Development, Yokohama, Japan.

Since 2009, he has been leading the development of ReRAM device and process technologies at Advanced Memory Development Center, Yokkaichi, Japan.



Satoru Takase received the B.E. and M.E. degrees in electrical engineering from Nagoya University, Nagoya, Japan, in 1988 and 1990, respectively.

He joined the Semiconductor Device Engineering Laboratory, Toshiba Corporation, Kawasaki, Japan, in 1990, where he designed high-speed circuitry for multibank DRAMs as a joint project with Rambus Inc., Mountain View, CA. In 1999, he presented a 1.6-GB/s 16-Bank Rambus DRAM in the IEEE International Solid State Circuits Conference (ISSCC).

From 2002 to 2006, he was in Austin, TX, as a member of joint development of a CELL Broadband Engine with Sony, Toshiba and IBM, where he designed circuits for an execution unit in a 64-bit Power PC Processor Element in 90-nm SOI technology, also contributed to the development of the power-performance management plan for 9-cores 65-nm CELL processor. In 2007, he returned to the Center for Semiconductor Research and Development, Toshiba Corporation, Yokohama, Japan, where he has been engaged in the design management for nonvolatile memory including NAND flash memory and ReRAM. He holds 43 issued U.S. patents and is author and coauthor of several technical papers.



Hirofumi Inoue received the B.S. and M.S. degrees from the Osaka University of Japan, Osaka, Japan, in 1988 and 1990, respectively.

After he joined Toshiba Corporation in 1990, he was engaged various leading edge memory development (DRAM/NAND) and memory module development. Since 2010, he joined JMDY (Toshiba-SanDisk Joint Memory Development, Yokkaichi, Japan) and develop 3D memory devices (include 3DNAND, 3DReRAM).



Luca Fasoli (M'95–SM'06) received the degree in electronic engineering and Ph.D. degree from the Politecnico di Milano, Milan, Italy, in 1993 and 1996, respectively. His Ph.D. research was focused on integrated devices and electronics for high-resolution X-ray spectroscopy and on high-sensitivity noise measurement systems.

In 1997, he joined Waferscale Integration (acquired by STMicroelectronics in 2000), where he was involved with the development and design of FLASH memory and FLASH-based PLDs for system-on-a-chip solutions. In 2001 he joined Matrix Semiconductor (acquired by Sandisk Corporation in 2006), where he was part of the team that produced multiple generations of nonvolatile 3-D storage devices. He is currently a Senior Director with Sandisk Corporation, Milpitas, CA, USA. In his role, he leads the product development of new-generation memories from concept definition to production. He is an author or coauthor of more than 25 technical publications and holds 77 U.S. patents.